

Thermodynamic Modeling of Directed Acyclic Trophic Graphs and Lindeman’s Efficiency Matrices in Biosphere Simulation Engines

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Sprint 001 – Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Simulating complex living systems requires strict adherence to physical conservation laws and thermodynamic constraints. In Sprint 001, we implemented Directed Acyclic Trophic Graphs (DATG) and Lindeman’s 10% Efficiency Rule matrices within `src/biosphere/trophic.ts`. By formalizing energy transfers through immutable monad state containers (`TrophicStateMonad`), our architecture guarantees the preservation of the First Law of Thermodynamics (mass-energy conservation) while correctly modeling entropic degradation and metabolic heat dissipation dictated by the Second Law. This paper outlines the theoretical framework, algorithmic implementations, and verification protocols established in Sprint 001.

1 Introduction and Theoretical Foundations

Ecosystem dynamics emerge from complex networks of energy acquisition, storage, and transfer. In computational biosphere simulations, failure to respect fundamental physical limits often results in runaway feedback loops, non-conservation of energy, or unphysical trophic amplification.

To resolve these challenges within the Web of Life engine, Sprint 001 establishes a rigorous thermodynamic framework rooted in systems ecology and nonequilibrium thermodynamics.

1.1 The First and Second Laws in Ecological Networks

1. **First Law of Thermodynamics (Mass-Energy Conservation):** Total system energy remains invariant under internal transformations. Energy entering via primary production must equal the sum of standing biomass accumulation, metabolic respiration, egested waste, and heat dissipation.
2. **Second Law of Thermodynamics (Entropic Degradation):** Energy transfers between trophic levels are inherently lossy. Following Lindeman’s foundational observations (1942), trophic efficiency ϵ is bounded between 1% and 15%, with the remainder degraded into unrecoverable metabolic heat.

2 Architectural Design & Implementation

The module `src/biosphere/trophic.ts` introduces explicit taxonomic nodes, directed edges representing consumption vectors, and an overarching trophic matrix manager.

2.1 Mass-Energy Conservation Invariants

For any trophic node i over a discrete time step Δt , the change in stored biomass energy (ΔB_i) is governed by:

$$\Delta B_i = \text{Input}_i - \text{Respiration}_i - \text{Egestion}_i - \text{ConsumptionLoss}_i + \text{Assimilation}_i \quad (1)$$

2.2 Lindeman's 10% Efficiency Matrix

The transfer function along any directed edge $e_{j \leftarrow i}$ from prey i to predator j is bounded by:

$$\text{EnergyTransferred}_{j \leftarrow i} = B_i \cdot \epsilon_{j \leftarrow i} \quad (2)$$

$$\text{MetabolicHeatLoss}_{j \leftarrow i} = B_i \cdot (1 - \epsilon_{j \leftarrow i}) \quad (3)$$

where $\epsilon_{j \leftarrow i} \in [0.01, 0.15]$, complying with empirical ecological limits.

3 Monadic State Transitions

To ensure deterministic, side-effect-free state propagation across ecosystem ticks, energy states are wrapped in the `TrophicStateMonad`, maintaining strict immutability and state traceability.

$$\sum \text{Biomass}_{\text{initial}} + \sum \text{Solar}_{\text{input}} = \sum \text{Biomass}_{\text{final}} + \sum \text{Heat}_{\text{dissipated}} \quad (4)$$

4 Verification and Results

Verification suites confirm that global system energy deltas equal zero across all integration steps when accounting for radiated heat, secondary transfer efficiencies satisfy $\epsilon \leq 0.15$, and cyclic dependency injections throw immediate structural validation errors.

For complete source code, test benches, and architectural RFCs, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.